



Flex Flow Tester 2.10

Interface Definition – User Manual

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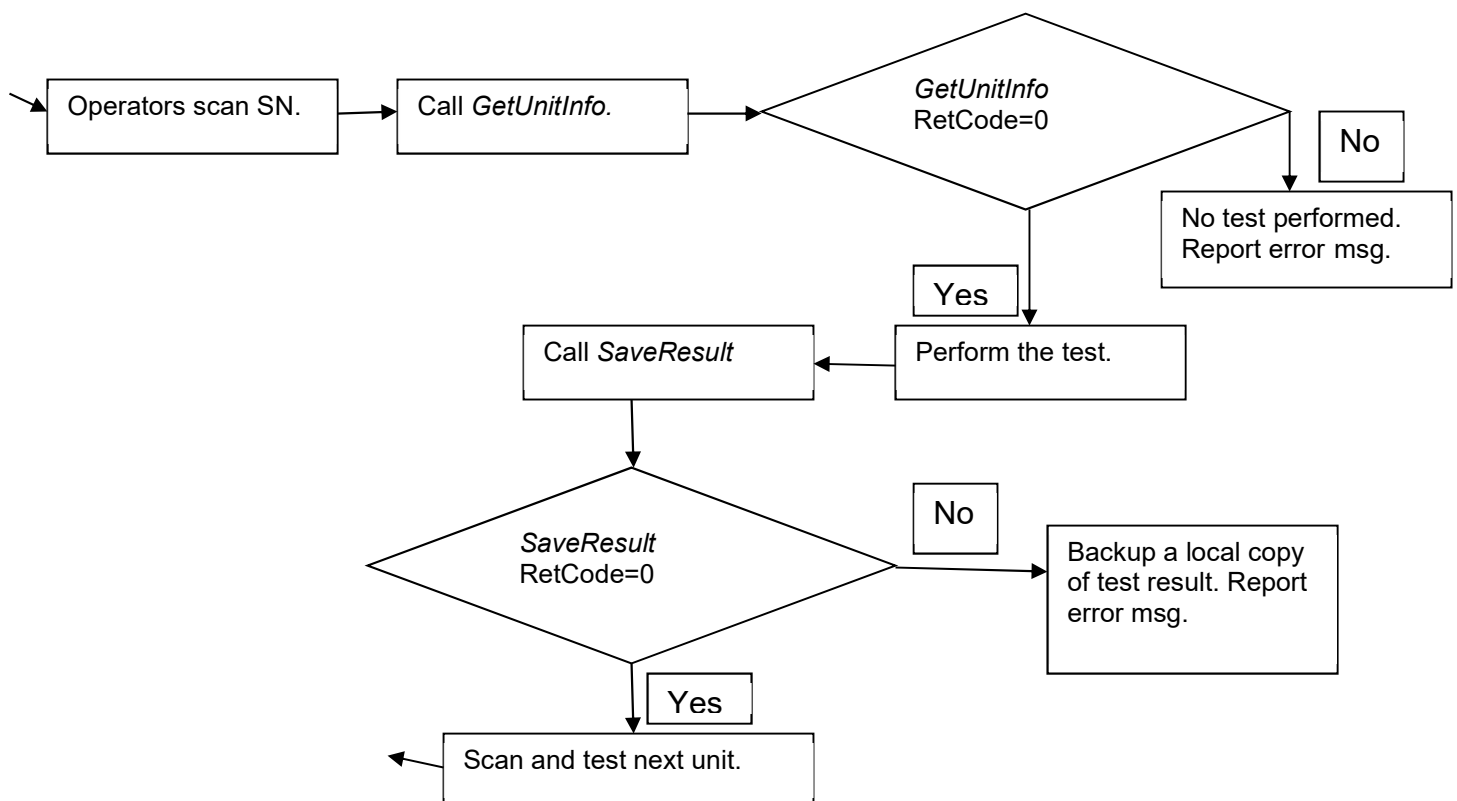
Overview

A test program can exchange data with Flexflow by either calling a .NET assembly installed on the same local machine, or calling a remote web service. We recommend non-Windows clients to use web service solution.

The interface provides two basic methods (*GetUnitInfo* and *SaveResult*) and one generic method (*Test Genric function*) for test programs to call when performing a test to device.

Test interface methods	When to call
<i>GetUnitInfo</i>	Before performing a test, test programs call <i>GetUnitInfo</i> to query unit information.
<i>SaveResult</i>	After performing the test, test programs call <i>SaveResult</i> to pass test status (<i>pass</i> or <i>fail</i>) and measurement data to Flexflow.

A typical control flow for test program:



1. Get Unit Info

GetUnitInfo takes two mandatory string parameters: Station Name and Serial Number . There are two optional parameters, UserID and Extra Data. If UserID is not set, then FFTester will not validate users against access right. Extra data just is necessary in case of we have a customize store procedure.

Request

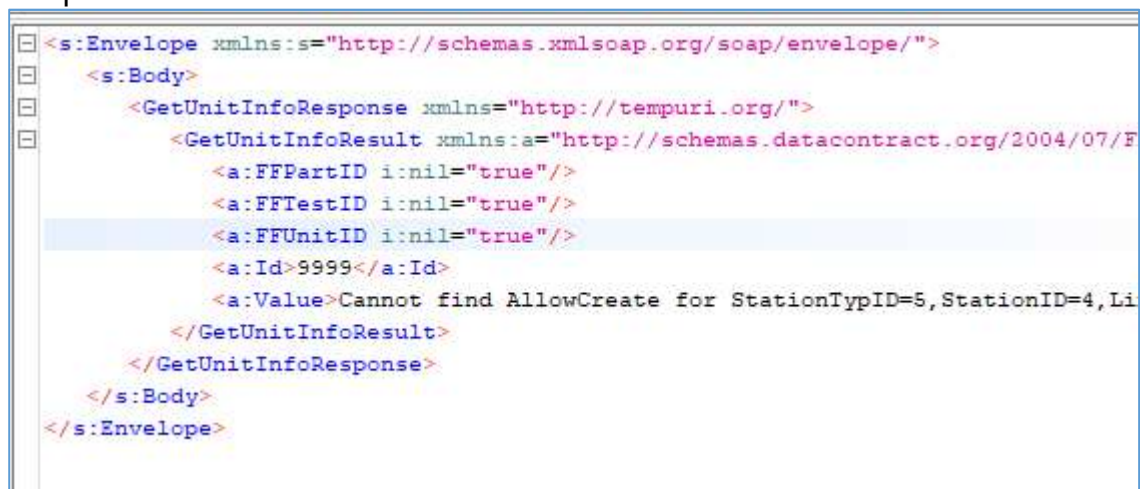


```

Request 1
http://qdlm02205:4450/
XML
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
  <soapenv:Header/>
  <soapenv:Body>
    <tem:GetUnitInfo>
      <!--Optional:-->
      <tem:serialNumber>FFTester1906190014</tem:serialNumber>
      <!--Optional:-->
      <tem:stationName>FFTester_AutoTest1</tem:stationName>
      <!--Optional:-->
      <tem:userId></tem:userId>
      <!--Optional:-->
      <tem:extraData></tem:extraData>
    </tem:GetUnitInfo>
  </soapenv:Body>
</soapenv:Envelope>

```

Response



```

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body>
    <GetUnitInfoResponse xmlns="http://tempuri.org/">
      <GetUnitInfoResult xmlns:a="http://schemas.datacontract.org/2004/07/F"
        <a:FFPartID i:nil="true"/>
        <a:FFTestID i:nil="true"/>
        <a:FFUnitID i:nil="true"/>
        <a:Id>9999</a:Id>
        <a:Value>Cannot find AllowCreate for StationTypID=5, StationID=4, Li
      </GetUnitInfoResult>
    </GetUnitInfoResponse>
  </s:Body>
</s:Envelope>

```

a. SOAP

Example of the Request:

--GetUnitInfo--Request

```
<soapenv:Envelope
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:tem="http://tempuri.org/">
  <soapenv:Header/>
  <soapenv:Body>
    <tem:GetUnitInfo>
      <!--Optional:-->
      <tem:serialNumber>FFTester1906190013</tem:serialNumber>
      <!--Optional:-->
      <tem:stationName>FFTester_AutoTest1</tem:stationName>
      <!--Optional:-->
      <tem:userId>admin</tem:userId>
    </tem:GetUnitInfo>
  </soapenv:Body>
</soapenv:Envelope>
```

Example of the response:

--GetUnitInfo--Response

```
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body>
    <GetUnitInfoResponse xmlns="http://tempuri.org/">
      <GetUnitInfoResult
xmlns:a="http://schemas.datacontract.org/2004/07/Flex.FFTester.Utilities"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance">
        <a:FFPartID i:nil="true"/>
        <a:FFTestID i:nil="true"/>
        <a:FFUnitID i:nil="true"/>
        <a:Id>0</a:Id>

        <a:Value><![CDATA[<UnitInfo><UnitData><Name>PartNumber</Name><Value>TestPart1</Value><UserID>admin</UserID></UnitData><UnitData><Name>StationID</Name><Value>FFTester_AutoTest1</Value><UserID>admin</UserID></UnitData><UnitData><Name>ProductionOrder</Name><Value>Test001</Value><UserID>admin</UserID></UnitData></UnitInfo>]]></a:Value>

      </GetUnitInfoResult>
```

```
</GetUnitInfoResponse>
</s:Body>
</s:Envelope>
```

b. REST

Example of the Request:

--GetUnitInfo— JSON Request

```
{
  "serialNumber": "string",
  "stationName": "string",
  "userId": "string"
}
```

Example of the response:

--GetUnitInfo—JSON Response

```
{
  "id": "string",
  "value": "string",
  "ffUnitId": "long",
  "ffTestId": "long",
  "ffPartId": "long"
}
```

2. Save Test Result

SaveResult takes one parameter, an XML string containing test results for Flexflow to save. Currently two formats of XML are supported, namely *Flexflow format* and *FTS format*. People must specify in FFTester.ini file which XML Schema (XSD) to use. FFTester.dll will validate the received test results against the chosen XSD, and then transform the test results into a tabular format before saving them into the SQL Server database.

Request

Request 1

http://qdlm02205:4450/

Raw XML

```
<?xml version="1.0" ?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:tem="http://schemas.xmlsoap.org/soap/extension/1.0/soap-enc" >
  <soapenv:Header/>
  <soapenv:Body>
    <tem:SaveResult>
      <!--Optional:-->
      <tem:ffTesterXmlRequest>
        <!--Optional:-->
        <tem:stationName>FFTester_AutoTest1</tem:stationName>
        <!--Optional:-->
        <tem:userId></tem:userId>
        <!--Optional:-->
        <tem:extraData></tem:extraData>
      </tem:SaveResult>
    </soapenv:Body>
  </soapenv:Envelope>
</?xml>
```


Response

```

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body>
    <SaveResultResponse xmlns="http://tempuri.org/">
      <SaveResultResult xmlns:a="http://schemas.datacontract.org/2004/07/Flex.FFTe
        <a:FFPartID>1</a:FFPartID>
        <a:FFTestID>74</a:FFTestID>
        <a:FFUnitID>1</a:FFUnitID>
        <a:Id>0</a:Id>
        <a:Value>Success</a:Value>
      </SaveResultResult>
    </SaveResultResponse>
  </s:Body>
</s:Envelope>

```

a. SOAP

Example of the Request:

This xml could be too large but a short example attached below.

```

<soapenv:Envelope
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:tem="http://tempuri.org/">
  <soapenv:Header/>
  <soapenv:Body>
    <tem:SaveResult>
      <!--Optional:-->
      <tem:ffTesterXmlRequest><![CDATA[<?xml version="1.0" ?>
<BATCH TIMESTAMP="2002-08-07 15:14:34.938991" SYNTAX_REV="1.1"
COMPATIBLE_REV="1.1">
<FACTORY NAME="" LINE="" TESTER="Autotest" FIXTURE="" SHIFT=""
USER="admin" />
<PRODUCT NAME="Phone One" REVISION="P1" FAMILY=""
CUSTOMER="" />
<REFS SEQ_REF="C:\FTS\Apps\Commodity GSM\Board_Level\Commodity
GSM Board Level Test PD3.seq"
FTS_REF="PM:v036,INT:V034(CFG:V013,RES:0.A9,MEA:1.14)" LIM_REF=""
CFG_REF="C:\FTS\Apps\Commodity GSM\Board_Level\Commodity GSM
Board Level Test PA1.cfg" CAL_REF="" INSTR_REF="" />
<PANEL ID="Undef" COMMENT="" RUNMODE="Debug"
TIMESTAMP="2002-08-07 15:14:35.924127" TESTTIME="120.107666"
WAITTIME="0.000000" STATUS="Passed">

```



```

<DUT ID="UnitSN001" COMMENT="" PANEL="0" SOCKET="1"
TIMESTAMP="2002-08-07 15:14:35.924127" TESTTIME="20.107666"
ENDTIME="2002-08-07 15:14:56.031793" STATUS="Passed">
  <GROUP NAME="Lid Place" STEPGROUP="Main" GROUPINDEX="1"
LOOPINDEX="1" TYPE="" RESOURCE="" MODULETIME="1"
TOTALTIME="1" TIMESTAMP="2002-08-07 15:14:49.481373"
STATUS="Passed">
    <TEST NAME="Lid Place" DESCRIPTION="Lid Place" UNIT=""
VALUE="" HILIM="" LOLIM="" STATUS="Passed" RULE="EQ" TARGET="" />
  </GROUP>
</DUT>
</PANEL>
</BATCH>]]></tem:ffTesterXmlRequest>
  <!--Optional:-->
  <tem:stationName>ENP-Cover-Autotest</tem:stationName>
  <!--Optional:-->
  <tem:userId>admin</tem:userId>
  <!--Optional:-->
  <tem:extraData></tem:extraData>
</tem:SaveResult>
</soapenv:Body>
</soapenv:Envelope>

```

Example of the response:

```

--Save Result— Response
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body>
    <SaveResultResponse xmlns="http://tempuri.org/">
      <SaveResultResult
xmlns:a="http://schemas.datacontract.org/2004/07/Flex.FFTester.Utilities"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance">
        <a:FFPartID>1</a:FFPartID>
        <a:FFTestID>26</a:FFTestID>
        <a:FFUnitID>1004</a:FFUnitID>
        <a:Id>0</a:Id>
        <a:Value>Success</a:Value>
      </SaveResultResult>
    </SaveResultResponse>
  </s:Body>
</s:Envelope>

```

b. REST

Example of the Request:

--Save Result— JSON Request

```
{
  "stationName": "string",
  "userId": "string",
  "batch": {
    "timestamp": "datetime",
    "syntaxRev": "string",
    "compatibleRev": "string",
    "factory": {
      "name": "string",
      "line": "string",
      "tester": "string",
      "fixture": "string",
      "shift": "string",
      "user": "string"
    },
    "product": {
      "name": "string",
      "revision": "string",
      "family": "string",
      "customer": "string"
    },
    "refs": {
      "seqRef": "string",
      "ftsRef": "string",
      "limRef": "string",
      "cfgRef": "string",
      "calRef": "string",
      "instrRef": "string"
    },
    "panel": {
      "id": "string",
      "comment": "string",
      "runMode": "string",
      "timestamp": "datetime",
      "testTime": "float",
      "waitTime": "float",
      "status": "string",
      "dut": [{
        "id": "string",
        "comment": "string",
        "panel": "string",
        "socket": "string",
        "timestamp": "datetime",
        "testTime": "float",
        "status": "string",
        "group": {
          "name": "string",
          "stepGroup": "string",
          "groupIndex": "integer",
          "loopIndex": "integer",
          "type": "string",
```

```

        "resource": "string",
        "moduleTime": "float",
        "totalTime": "float",
        "timeStamp": "string",
        "status": "string",
        "extra": {
            "groupextra": "string"
        },
        "groups": [
            {
                "name": "string",
                "stepGroup": "string",
                "groupIndex": "integer",
                "loopIndex": "integer",
                "type": "string",
                "moduleTime": "float",
                "totalTime": "float",
                "timestamp": "string",
                "status": "string",
                "tests": [
                    {
                        "name": "string",
                        "description": "string",
                        "unit": "string",
                        "value": "string",
                        "hilim": "string",
                        "lolim": "string",
                        "status": "string",
                        "rule": "string",
                        "target": "string"
                    }
                ]
            },
            {
                "name": "string",
                "stepGroup": "string",
                "groupIndex": "integer",
                "loopIndex": "integer",
                "type": "string",
                "moduleTime": "float",
                "totalTime": "float",
                "timestamp": "datetime",
                "status": "string"
            }
        ]
    },
    "extra": {
    }
}

```

Example of the response:

--Save Result— JSON Response

```
{
  "id": "string",
  "value": "string",
  "ffUnitId": "long",
  "ffTestId": "long",
  "ffPartId": "long"
}
```

3. Generic Function.

The Generic Function method is being used for a customized step on the FF Autotest stations such as calling an UDP or a customized DLL and perform different kind of validations/steps before to record the result of the test.

This function allows run a customize process to execute your own tests.

@FunctionName: is the name of customize procedure

@SPInputParameter: is the data received in the request parameter called splInputParameter

@StationName: is the incoming station, also received within the request

@UserID: is the user id from ffEmployee table. This indicates which user is performing the request.

@Output: is an output parameter that can be filled within the stored procedure and retrieved as part of response

@ErrorDescription: another output parameter which is also visible within the response.

a. SOAP

Example of the Request:

--GenericFunction— Request

```
<soapenv:Envelope
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:tem="http://tempuri.org/">
  <soapenv:Header/>
  <soapenv:Body>
    <tem:ExecuteGenericFunction>
      <!--Optional:-->
      <tem:functionName>GenericFunctionName</tem:functionName>
      <!--Optional:-->
    <tem:splInputParameter>InputParametersInXmlFormat</tem:splInputParameter>
  </soapenv:Body>
</soapenv:Envelope>
```

```

    <!--Optional:-->
    <tem:stationName>StationName</tem:stationName>
    <!--Optional:-->
    <tem:userId>Username</tem:userId>
    <!--Optional:-->
    <tem:output></tem:output>
    <!--Optional:-->
    <tem:errorDescription></tem:errorDescription>
  </tem:ExecuteGenericFunction>
</soapenv:Body>
</soapenv:Envelope>

```

Example of the response:

--GenericFunction— Response

```

<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/">
  <s:Body>
    <ExecuteGenericFunctionResponse xmlns="http://tempuri.org/">
      <ExecuteGenericFunctionResult
xmlns:a="http://schemas.datacontract.org/2004/07/Flex.FFTester.Utilities"
xmlns:i="http://www.w3.org/2001/XMLSchema-instance">
        <a:FFPartID i:nil="true"/>
        <a:FFTestID i:nil="true"/>
        <a:FFUnitID i:nil="true"/>
        <a:Id>0</a:Id>
        <a:Value>returned value</a:Value>
      </ExecuteGenericFunctionResult>
      <output>Response Output Message from Stored Procedure +
DLL</output>
      <errorDescription></errorDescription>
    </ExecuteGenericFunctionResponse>
  </s:Body>
</s:Envelope>

```

b. REST

Example of the Request:

--GenericFunction— JSON Request

```
{
  "functionName": "string",
  "spInputParameter": "string",
  "stationName": "string",
  "userId": "string"
}
```

Example of the response:

--GenericFunction— JSON Response

```
{
  "id": "string",
  "value": "string",
  "ffUnitId": "long",
  "ffTestId": "long",
  "ffPartId": "long"
}
```

4. XML Cache File Format

The output cache files between FTS_RES.DLL and the external hook applications are a major interface for the FTS framework. Once this interface is defined and taken into use, it should not be radically changed in a way that would make old hook applications not work.

XML file format (“eXtensible Markup Language”) is used as the basis for the output cache files. It defines the general syntax but any specifics are down to the FTS tags as defined below. The best way to describe this is possibly to start by a sample cache file and then go through the individual tag names one by one...

```
<?xml version="1.0" ?>
<BATCH
  TIMESTAMP="..."
  SYNTAX_REV="..."
  COMPATIBLE_REV="..."
  COMMENT="...">

  <FACTORY
    NAME="..."
    LINE="..."
```

```

TESTER="..."
FIXTURE="..."
SHIFT="..."
TESTCATEGORY=""
SYSTEMTYPE=""
USER="..." />
<PRODUCT
  NAME="..."
  REVISION="..."
  FAMILY="..."
  CUSTOMER="..." />
<REFS
  SEQ_REF="..."
  FTS_REF="..."
  APP_REF="..."
  LIM_REF="..."
  CFG_REF="..."
  CAL_REF="..."
  INSTR_REF="..." />
<PANEL
  ID="..."
  COMMENT="..."
  RUNMODE="..."
  TIMESTAMP="..."
  TESTTIME="..."
  WAITTIME="..."
  STATUS="">
  <DUT
    ID="..."
    COMMENT="..."
    PANEL="..."
    SOCKET="..."
    TIMESTAMP="..."
    TESTTIME="..."
    STATUS="...">
    <GROUP
      NAME="..."
      STEPGROUP="..."
      GROUPINDEX="..."
      LOOPINDEX="..."
      TYPE="..."
      RESOURCE="..."
      MODULETIME="..."
      TOTALTIME="..."
      TIMESTAMP="..."

```



```

        STATUS="...">
        <TEST
            NAME="..."
            DESCRIPTION="..."
            UNIT="..."
            VALUE="..."
            HILIM="..."
            LOLIM="..."
            STATUS="..."
            RULE="..."
            TARGET="..."
            DATATYPE="..." />

        <GROUP> (Sub Groups) </GROUP>

    </GROUP>

    <GROUP> (More groups) </GROUP>

</DUT>

<DUT> (More duts) </DUT>

</PANEL>

<PANEL> (More panels) </PANEL>

</BATCH>

```

This same syntax can be used for both PANEL files (containing data for many DUTs) and DUT files. In the DUT files, there still needs to be a panel wrapper block around the DUT block, but that is no problem. We could even think of putting several panel blocks in the same file (e.g. to collect all results of a whole day or work shift together) but that is not required in the temporary cache files. (It may be done for permanent XML storage, though, by some hook application.)

Back to the issue...

Let's go through the sample once more, one tag at a time.

BATCH element

The BATCH element groups all PANELS that is included in the file together with some factory and test system related information. The BATCH element may contain one FACTORY, one ELEMENT, one REFS element, and one or more PANEL elements.

Attribute	Type	Max size	Explanation
<i>General:</i>			
SYNTAX_REV	text	(not to data-base)	Revision of FTS cache file syntax used (e.g. “1.2”).
COMPATIBLE_REV	text	(not to data-base)	Revision that this file is compatible with (e.g. “1.1”). As long as only new attributes or elements are added, new files are compatible. Individual hooks that depend on attributes or elements does not affect this attribute.
TIMESTAMP	date+time (automatic)	N/A	Date and time (including time zone) when the product was tested. Sample contents: “2001-10-11 18:44:14.01232345” This is the test stations local time.
COMMENT	text (optional)	(tbd)	Free text comments inserted by the user (e.g. description of failing hardware, remark of changed equipment, weather etc...)

FACTORY element:

The FACTORY element describes information about the factory and factory set up when the test took place.

Attribute	Type	Max size	Explanation
NAME	text	(tbd)	Name or code of the manufacturing site (e.g. “PL/TCW”, “SE/VAS”, “FI/KYR”).
LINE	text	(tbd)	Production line ID within the factory (e.g. “HFU2-4”)
TESTER	text	(tbd)	Tester hardware ID (e.g. “KYR-C01432”) Usually the machine name of the PC in the tester.
FIXTURE	text (optional)	(tbd)	Identity of the fixture that was used to test the panel. Only needed when a tester has more than one independent fixtures.
SHIFT	text (optional)	(tbd)	Work shift or team ID that the operator belongs to
USER	text	(tbd)	ID of the person using the tester (e.g. TestStand login name)
TESTCATEGORY	text (optional)	(tbd)	Test category is the type of test that were performed. Useful when a product is tested more than once, with different programs in the same test

			station.
SYSTEMTYPE	text (optional)	(tbd)	The system type. All test stations with the same type are identical.

PRODUCT element:

The PRODUCT element keeps information about the product that was tested. This element may occur in the top BATCH level, if all products test were tested was the same, but it may also be specified within the PANEL or DUT level if different product types are included in the file.

Attribute	Type	Max size	Explanation
NAME	text	(tbd)	Name or codename of the product.
REVISION	text	(tbd)	Product revision (e.g. changed PCB layout, processor model, ...)
FAMILY	text (optional)	(tbd)	Name or codename of the product family (if any).
CUSTOMER	text (optional)	(tbd)	Name or codename of the customer (if any).

REFS element:

The REFS element holds information about the application that was used to perform the test. It also informs about the configuration for the test system, such as program revisions, instrumentation and calibration.

Attribute	Type	Max size	Explanation
FTS_REF	text (automatic)	(tbd)	FTS system reference (distribution version & release date?)
SEQ_REF	text	(tbd)	Test application reference (program code version & date)
APP_REF	text (optional)	(tbd)	Optional application reference other than SEQ_REF, LIM_REF, CFG_REF.
LIM_REF	text	(tbd)	Limits file reference (version & date)

CFG_REF	text	(tbd)	Configuration files reference (version & date)
CAL_REF	text (optional)	(tbd)	Calibration reference (person name & calibration date)
INSTR_REF	text (optional)	(tbd)	Instrument reference (serial numbers of key instruments etc.)

PANEL element:

The PANEL element identifies a panel that were tested. If the DUT are single entities and no panels were used, this element will just be a wrapper for the DUT. The PANEL element also contains some runtime information about the test execution. The PANEL contains at least one DUT element and may contain a PRODUCT element.

Attribute	Type	Max size	Explanation
ID	text (optional)	(tbd)	Identifier (e.g. serial number) for the panel
TIMESTAMP	date+time (automatic)	N/A	Date and time (including time zone) when the product was tested. Sample contents: "2001-10-11 18:44:14.01232345" This is the test stations local time.
RUNMODE	text	(tbd)	The run mode that were used during test "PRODUCTION", "DEBUG", "GOLDEN", "SELFTEST"... This is used to tag the data in the database, so that analysis can be made only on production tests. We don't want to include debugging sessions in the test yield for example.
WAITTIME	float	N/A	Time spent waiting for the plate to arrive (from end of previous TESTTIME to the beginning of this one)
TESTTIME	float	N/A	Total testtime of the plate in seconds (from start of first DUT to the end of the last)
COMMENT	text (optional)	(tbd)	Free text comments inserted by the user (e.g. description of failing hardware, remark of changed equipment, weather etc...)
STATUS	text	(tbd)	"Passed" / "Failed" / "Terminated" / "Aborted" / "Error" / "Done" / "Skipped" / "Intermediate" / "Planned"

DUT element:

The DUT element identifies the DUT that were tested, including which panel position it has (if panel mounted) and in which test socket were used. The test that was performed is represented in one or many the underlying GROUP elements.

Attribute	Type	Max size	Explanation
ID	text	(tbd)	Identifier (e.g. serial number) for the Device Under Test
TIMESTAMP	date+time (automatic)	N/A	Date and time (including time zone) when the product was tested. Sample contents: "2001-10-11 18:44:14.01232345" This is the test stations local time.
PANEL	integer (optional)	N/A	Index of the DUT within the panel PCB (1...N). Can be left out if there is only one DUT per panel.
SOCKET	integer	N/A	Number of test socket (0...N-1 where the DUT got tested).
TESTTIME	float	N/A	Seconds it took to test this DUT.
STATUS	text	(tbd)	"Passed" / "Failed" / "Terminated" / "Aborted" / "Error" / "Done" / "Skipped" / "Intermediate" / "Planned"

GROUP element:

The GROUP element is for grouping TEST elements and also used for reflecting the test program architecture. A GROUP element may contain TEST or GROUP elements. A GROUP element represents the logical grouping that was used in the test program. This can be either a code module that were executed or the test executives grouping feature. In both case, the GROUP element represents something that were executed.

Attribute	Type	Max size	Explanation
NAME	text	(tbd)	Short name for the group of tests. Must NOT be changed once production has started!
DESCRIPTION	text (optional)	(tbd)	Longer name shown to the user and placed in reports. This field is not used for automatic processing and may therefore be changed.
STEPGROUP	text (optional)		If the test executive identifies parts of a GROUP, such as Setup, Main or Cleanup, is indicated in STEPGROUP
GROUPINDEX	integer		Represents the index of this GROUP within its parent

	r		GROUP
LOOPINDEX	integer		If the group is a looping GROUP, this is the loop iteration. If the GROUP is not looping or the group represents the cumulative result for a looping GROUP the LOOPINDEX = -1
TYPE	text		A name representing the type of the GROUP that were executed.
RESOURCE	text (optional)		A textual description of the resource that were executed, such as the name of the VI, function or sequence.
MODULETIME	float		The time it took to execute the code module, excluded overhead for verification, setup, pre and post actions.
TOTALTIME	float	N/A	Total testtime of the group of tests in seconds
TIMESTAMP	date+ time (automatic)	N/A	Date and time (including time zone) when the product was tested. Sample contents: "2001-10-11 18:44:14.01232345" This is the test stations local time.
STATUS	text	(tbd)	"Passed" / "Failed" / "Terminated" / "Aborted" / "Error" / "Done" / "Skipped" / "Intermediate" / "Planned"
COMMENT	text (optional)	(tbd)	Free text comments inserted by the user (e.g. description of failing hardware, remark of changed equipment, weather etc...)

TEST element:

The TEST element contains the actual value that is verified and its limits.

Attribute	Type	Max size	Explanation
NAME	text	(tbd)	Short name (ID) for the test Must NOT be changed once production has started!
DESCRIPTION	text (optional)	(tbd)	A more informative name for the test
STATUS	text	(tbd)	"Passed" / "Failed" / "Terminated" / "Aborted" / "Error" / "Done" / "Skipped" / "Intermediate" / "Planned"
VALUE	text	(tbd)	Measured data without unit; just raw data
LOLIM	text	(tbd)	The low limit in the comparison. If a unary comparison ("GT", "LE") this is the limit that the value were

			compared to.
HILIM	text (optional)	(tbd)	The high limit in the comparison. If a unary comparison ("GT", "LE") HILIM is not used.
RULE	text	(tbd)	Comparison rule used in the test ("LT"/"GT"/"LE"/...)
TARGET	text (optional)	(tbd)	Lower limits (without unit part)
DATATYPE	text	(tbd)	Specifies data type of Value, Lo/Hi-Lim and Target
UNIT	text	(tbd)	The unit of the measured value.

ERROR element

If an error occurs, it can be correct to report this error into the XML. In those cases the ERROR element is used. Where the error occurred is indicated by the elements position in the XML

Attribute	Type	Max size	Explanation
DESCRIPTION	text	(tbd)	A clear text description of the runtime error
VALUE	integer	(tbd)	A numeric value describing the error

EXTRA element

In addition to the built-in 'standard' fields described above the applications may also attach their own proprietary fields into the XML files. The EXTRA element does not specify which attributes to use.

The EXTRA element may occur within the BATCH, PANEL, DUT and GROUP elements.

Attribute	Type	Max size	Explanation
<ANY>	text	(tbd)	Any attribute name is valid. Its up to the test engineer to set up and identify the content of the <EXTRA> element.

5. XML cache file Semantics

Even though the XML syntax is specified, it is not quite obvious how it is used. Dependent on the test executive the XML might be used in different ways.

This chapter defined how FTS uses the XML and defines the data that goes into different attributes.

- The BATCH element wraps “all panels that were started at the same time”.
- The PANEL element is not used in FTS more than as a placeholder for DUTs and the RUNMODE.
- The DUT element represents the DUT that were tested.
- The GROUP element represents the execution of a sequence or a code module. Actually it represents a step. A sequence that is executed is initiated from a “SequenceCall Step”
- The TEST represents the values that the step used for evaluating the status expression

Example:

One GROUP and one underlying TEST element will represent a test step that uses the NumericLimitTest steptype.

```
<GROUP
  NAME="Numeric Limit Test"
  STEPGROUP="Main"
  GROUPINDEX="3"
  LOOPINDEX="-1"
  TYPE="NumericLimitTest"
  RESOURCE=""
  MODULETIME="0.000000"
  TOTALTIME="0.000210"
  TIMESTAMP="2002-09-12 06:57:16.335747"
  STATUS="Passed">

  <TEST NAME="Numeric Limit Test" DESCRIPTION="" UNIT=""
    VALUE="15"
    HILIM="100"
    LOLIM="0"
    STATUS="Passed"
```



```

    RULE="GELE" TARGET=""
    DATATYPE="Number" />
</GROUP>

```

The BATCH will represent all PANELs (DUTs) that were started at the same time. In this version of FTS, we only have one Asynchronous Parallel execution model that means that only one DUT is started at once. So only one PANEL will exist in the BATCH. But in the next version, we will have BATCH support in FTS, that means that many PANELs will be started simultaneously.

Example:

A burn-in chamber is loaded with 100 units. Each run in its own test socket (in parallel) but they are all started at once, with one button press and the test is finished when all sockets are tested.

The DUT element will contain information about the tests that were executed during the BATCH. Normally, when the user presses the start button, the test is executed once and the DUT element will contain one underlying GROUP element.

Example:

As the implementation is today, the NAME attribute of this test is taken from the sequence call step name. This step is located in the process model and is called “MainSequence Callback”. This might look weird, as it cannot be located anywhere in the application sequence. But the RESOURCE attribute contains “MainSequence” and is the sequence that is executed.

```

<GROUP
  NAME="MainSequence Callback"
  STEPGROUP="Main"
  GROUPINDEX="38"
  LOOPINDEX="-1"
  TYPE="SequenceCall"
  RESOURCE="MainSequence"
  MODULETIME="1.480148"
  TOTALTIME="1.480853"
  TIMESTAMP="2002-09-12 06:57:15.880882"
  STATUS="Passed">

```

(Left out groups and tests ...)

In TestStand, it is possible to let the sequencer loop on a step until a certain condition is met. This condition can be “Loop until pass then consider the test passed” or “Loop 157 times and then consider test passed if 37% failed”. In this looping case, one GROUP element with LOOPINDEX = -1 and without any first level TEST elements is inserted. This element tells the status of the test and the cumulative time for the loop. Inside this group, the iterations are inserted, where each GROUP contains all results from each iteration. The LOOPINDEX is set to 0, 1, 2, ..., n-1 and each GROUP contains TEST (or GROUP) elements. In this way we keep all measurements from all iterations but we don't need to process them into the database. If we chose not to record results for each iteration (option in TestStand), only an empty cumulative GROUP is present that represents the status of the test.

Example:

```
<GROUP NAME="Numeric Limit Test_2" LOOPINDEX="-1" STATUS="Passed">
  <GROUP NAME="Numeric Limit Test_2" LOOPINDEX="0" STATUS="Failed">
    <TEST NAME="Numeric Limit Test_2" VALUE="0" HILIM="5"
    LOLIM="2"
      STATUS="Failed" RULE="GELE" DATATYPE="Number"/>
  </GROUP>

  <GROUP NAME="Numeric Limit Test_2" LOOPINDEX="1" STATUS="Failed">
    <TEST NAME="Numeric Limit Test_2" VALUE="1" HILIM="5" LOLIM="2"
      STATUS="Failed" RULE="GELE" TARGET=""
      DATATYPE="Number"/>
  </GROUP>

  <GROUP NAME="Numeric Limit Test_2" LOOPINDEX="2" STATUS="Passed">
    <TEST NAME="Numeric Limit Test_2" VALUE="2" HILIM="5" LOLIM="2"
      STATUS="Passed" RULE="GELE" DATATYPE="Number"/>
  </GROUP>
```

(Groups left out ...)

```
<GROUP NAME="Numeric Limit Test_2" LOOPINDEX="9" STATUS="Failed">
  <TEST NAME="Numeric Limit Test_2" VALUE="9" HILIM="5" LOLIM="2"
    STATUS="Failed" RULE="GELE" TARGET="" DATATYPE="Number"
  />
</GROUP>

</GROUP>
```

Each step, by default, contains a ReportText sub property. This content is intended to go into the report in some way. So, FTS takes the value of the sub property and inserts it as an EXTRA element inside the containing group.

Example:

```
<EXTRA ReportText="This is written in the Step.Result.ReportText" />
```

Runtime errors can occur in a sequence. This is when a step executes and the Step.Result.Error.Occurred flag is True. In these cases, the STATUS attribute of the GROUP is “**Error**” and no TEST element or GROUP elements are beneath the GROUP. Instead an ERROR element is inserted in the GROUP.

Example:

```
<GROUP NAME="Demonstrate RunTimeError" STATUS="Error">
  <ERROR DESCRIPTION="A runtime error occurred" VALUE="42" />
</GROUP>
```

The last part of this chapter describes what data goes into respective property of the different elements.

TEST element:

Attribute	Explanation
NAME	Name of the TEST. Usually the same as the step name, but it can differ, for example in the MultipleNumericLimit test, separate names can be set for the different values.
DESCRIPTION	The description of a step is taken from the step, if a lookup for it is provided, otherwise it is taken from the comment of the step
STATUS	The status is transparently taken from TestStand, that means that it will be either “Passed” or “Failed”. It may happen that TestStand reports other statuses, especially for custom steps, Those are transparently inserted to the XML.
VALUE	Measured data without unit; just raw data taken directly from the step.
LOLIM	The limits are taken directly from the step
HILIM	The limits are taken directly from the step
RULE	The comparison rule is one of the one that TestStand

	prescribes GE, GT, LE, LT, EQ, NE, GTLT, GTLE, GELT, GELE, LIMITS, MATCH For the NONE case, no TEST element will appear in the XML.
TARGET	Not used
DATATYPE	Will be one of “String”, “Boolean”, “Number” or “FtsWaveform_t”
UNIT	Taken directly from the step, if any.

GROUP element

Attribute	Explanation
NAME	Is the name of the calling step. For the SequenceCall Step, it is still the name of the step, the sequence that is executed is indicated in the RESOURCE attribute
DESCRIPTION	No test description is used
STEPGROUP	Will be the TestStand step group in which the step were located “Setup”, “Main”, “Cleanup”
GROUPINDEX	The index of the step within its sequence
LOOPINDEX	-1 for non looping or cumulative result of an executed step. 0-n for each iteration of a looping step.
TYPE	The step type of the executing step, For example NumericLimitTest
RESOURCE	A string that describes what code module (vi, function in dll, sequence in sequence file) were executed.
MODULETIME	The time it took to execute the code module, excluded overhead for verification, setup, pre and post actions.
TOTALTIME	Total testtime of the group of tests in seconds
TIMESTAMP	Date and time (including time zone) when step were executed expressed in ISO8601 format. Sample contents: “2001-10- 11T18:44:14.01232345+01:00”
STATUS	Will be the status for the executed step. This is transparently copied from TestStand and will usually be one of the following “Passed”, “Failed”, “Done”, “Error”. Other statuses may occur if a custom step defines another.

COMMENT	Free text comments inserted by the user form the FTS UI
---------	---

DUT element

Attribute	Explanation
ID	The serial number for the DUT that was entered in the user interace.
TIMESTAMP	Date and time (including time zone) when step were executed expressed in ISO8601 format. Sample contents: "2001-10-11T18:44:14.01232345+01:00"
PANEL_POS	As FTS don't have PANEL support, this is always set to 0
SOCKET	The test socket index. This starts with 0 for the first socket and up to N-1
TESTTIME	Seconds it took to test this DUT to be tested.

PANEL element

Attribute	Explanation
ID	Not used
TIMESTAMP	Date and time (including time zone) when step were executed expressed in ISO8601 format. Sample contents: "2001-10-11T18:44:14.01232345+01:00"
RUNMODE	The run mode that were used during test "PRODUCTION", "DEBUG", "GOLDEN", "SELFTEST"...
WAITTIME	Not used, set to 0
TESTTIME	Same as for the DUT
COMMENT	Not used

BATCH element

Attribute	Explanation
SYNTAX_REV	Defined by RES subsystem on format "1.2"
COMPATIBLE_REV	Defined by RES subsystem on format "1.1"
TIMESTAMP	Date and time (including time zone) when step were executed expressed in ISO8601 format. Sample contents: "2001-10-

	11T18:44:14.01232345+01:00"
COMMENT	Not used

FACTORY element:

Attribute	Explanation
NAME	Name or code of the manufacturing site (e.g. "PL/TCW", "SE/VAS", "FI/KYR"). The name should be something like "<Country>/<City>". Does an Flextronics standard exist? Specified in the [StationOptions] section in the configuration subsystem.
PRODLINE	Production line ID within the factory (e.g. "HFU2-4") Specified in the [StationOptions] section in the configuration subsystem.
TESTER	Tester hardware ID (e.g. "KYR-C01432") Specified in the [StationOptions] section in the configuration subsystem.
FIXTURE	Identity of the fixture that was used to test the panel. Usually set during test program load. Can be identified during runtime by reading fixture ID from the fixture and added to the system settings.
SHIFT	Work shift or team ID that the operator belongs to. Specified in the [StationOptions] section in the configuration subsystem, unless defined during runtime.
USER	ID of the person using the tester. This is the TestStand login name
SYSTEMTYPE	Taken from the test station configuration [StationOptions] SystemType
TESTCATEGORY	Information obtained during test program loading.

PRODUCT element:

Attribute	Explanation
NAME	Name of the product specified during test program load. Usually from the program database.
REVISION	Name of the product specified during test program load. Usually from the program database.
FAMILY	Name of the product specified during test program load. Usually from the program database.
CUSTOMER	Name of the product specified during test program load. Usually from the program database.

REFS element:

Attribute	Explanation
FTS_REF	FTS system reference. This attribute lines up the versions of all major subsystems in that span the FTS system. "PM:v043,INT:V037 (CFG:V1.0, RES:1.01, LIM:V1.0)"
APP_REF	The full path of the sequence file
LIM_REF	The full path of the limit file
CFG_REF	The full path of the config file
CAL_REF	Not used
INSTR_REF	Not used

6. Validation

For validation of an FTSRES-XML, XML Schema Definition files (.xsd) has been developed.

FTSRes-v1.2.xsd defines version 1.2 of the FTSRES-XML.



FTSRes-v1.2.xsd

FTSRes-v1.3.xsd defines version 1.3 of the FTSRES-XML.



FTSRes-v1.3.xsd

FTSRes-v1.4.xsd defines version 1.4 of the FTSRES-XML.

<XD>

FTSRes-v1.4.xsd

7. Error Messages

For GetUnitInfo and SaveResult, it will return 0 for success. For failure, these error codes will be returned.

120110005	NextStateID must be numeric.
120110010	The serial numbers should not be matched to multiple UnitID.
120110050	Cannot take NULL field value for nonexistent unit.
120111010	Cannot convert UnitID into integer.
180020100	Cannot find StatusID in luProductionOrderStatus table.
180020600	Update to ffProductionOrder table has failed.
180020700	Production order does not exists in ffProductionOrder table.
180020800	Insertion to ffProductionOrderHistory table has failed.
180020810	Failed to update ffLineOrder to set Priority to zero.
180020830	Failed to update ffLineOrder to shift priority of line orders.
180030001	Missing infromatin in the PackageSettingsXMLstring.
180030050	Production order does not exist in ffProductionOrder table.
180030060	@Name is not supplied.
180030080	@Name is not defined in luProductionOrderDetailDef table.
180030090	Value does not exist in ffPartDetail table.
180040001	Error updating ffUnit..
180040200	UnitSerianumber not found in DB.
180040300	The unit does not belong to any production order.
180040400	Invalid Production Order.
180040500	The production order has incompatible status.
180040600	Invalid Line ID.
180040700	Cannot add the quantity because it would exceed the allowed line quantity.
180040800	Failed to upate ffLineOrder table with quantity.
180040900	Some parameters are not filled out correctly (DW) 26.08.04.
180050001	Cannot find ID for 'Active' in luProductionOrderStatus table.
180050002	Cannot find ID for 'Released' in luProductionOrderStatus table.
180050003	The line has not defined a highest priority production order.
180050004	The line production order has not defined a part ID.
230010010	Incoming UnitStateID is NULL.
230010020	Incoming StationTypeID is NULL.
230010030	Routing error (entered a wrong station).
230010040	Unable to find a RouteID.
230010045	The unit is in scrap or hold status.
230010050	The station does not allow NULL incoming state.

250010010	Incoming UnitStateID is NULL.
250010020	Incoming StationTypeID is NULL.
250010030	Cannot find StateDescription in luOutputStateDef table.
250010040	Unable to find a RouteID.
250010050	Cannot resolve multiple next unit state for current station.
250010070	Cannot find next outgoing unit state for current station.
270001000	Some group entry points to nonexistent parent in @GROUP.
270001100	Parent cannot point to itself in @GROUP.
270001200	Some group entry points to nonexistent @DUT.
270001300	Some test entry points to nonexistent @DUT.
270001400	Some test entry points to nonexistent @GROUP.
270001410	Some TextExtra entry has no parent.
270001420	Some GroupExtra entry has no parent.
270001430	Some DUTExtra entry has no parent.
270001500	There are some rows in #GROUP not having parent.
270001600	Cannot find unit serial number in ffUnit table.
270001700	Invalid station description in TESTER attribute of FACTORY element.
270001710	Cannot find LineID for the given station.
270001890	Cannot insert to ffTestRef table.
270001900	One of the DUT in panel has routing error.
270001910	StartToPO is not defined for basic function frmTestSaveResult.
270001920	ChangePartNumber is not defined for basic function frmTestSaveResult.
270001930	Failed to find reverse part number when downgrade part number.
270001940	CompleteToPO is not defined for basic function frmTestSaveResult.
270001950	Cannot insert to luJig table.
270002000	SQL Server failed to insert into ffTestPanel table.
270002100	SQL Server failed to insert into ffTestPanelExtra table.
270002300	SQL Server failed to insert into ffTest table.
270002400	SQL Server failed to insert into ffTestExtra table.
270002500	SQL Server failed to update @DUT table.
270002590	SQL Server failed to insert into ffTESTSTEPPASS table.
270002600	SQL Server failed to update @GROUP table.
270002700	SQL Server failed to insert into ffTestStepPassExtra table.
270002800	SQL Server failed to insert into ffMeasurementDetailPass table.
270002900	SQL Server failed to update @TEST table.
270003000	SQL Server failed to insert into ffMeasurementDetailPassExtra table.
270003100	SQL Server failed to update @GROUP table.
270003200	SQL Server failed to insert into ffTestStepFailExtra table.
270003300	SQL Server failed to insert into ffMeasurementDetailFail table.
270003400	SQL Server failed to update @TEST table.
270003410	SQL Server failed to insert into ffMeasurementDetailFailExtra table.
270003500	SQL Server failed to update ffUnitDetail table.
270003505	Test value cannot be trivial when it is going to be saved into ffSerialNumber.
270003510	SQL Server failed to insert ffSerialNumber table.
270003515	SQL Server failed to update ffSerialNumber table.
270003520	The DUT already has a different value in ffSerialNumber.
270003521	SQL Server failed to insert ffSerialNumber table.
270003600	SQL Server failed when call uspEXTTestSaveResult.
280000026	Cannot resolve PartID.
280000027	The input @PartID is not equal to PartID from ffImportProduct.

280000029	The input @PartID is not equal to PartID from ffUnit.
280000030	Unit already exists. Cannot create it again..
280000031	The input @PartID is not equal to PartID from ffImportProduct.
280000034	Cannot find PartID because @PartID is NULL, @ProductionOrderID is null and no info from ffImportProduct and ffUnit.
280000035	Invalid RMA Number being received.
280000036	The input @PartID is not equal to PartID from ffUnit.
280000037	RMA already close. Please select a new RMA#.
280000038	Cannot get PartID.
280000045	RMA number does equal the one in ffImportProduct table. 03/16/2005 added @AllowLoopers=1.
280000050	Cannot find incoming unit state id for the current station.
280000051	Unit serial number pattern check failed.
280000060	SQL Server failed to insert unit into ffUnit table.
280000070	SQL Server failed to insert unit into ffUnitDetail table.
280000080	SQL Server failed to update ffUnit table for RMA project type.
280000083	Failed to insert failure code into ffTest table.
280000084	Failed to insert record into ffTest table.
280000085	For a loopier the unit StatusID must be 1 when it comes back.
280000090	SQL Server failed to update ffUnitDetail for RMA project type.
280000091	Failed updating ffImportProduct.
280000092	Failed updating ffImportProduct.
280000100	Failed to insert ffHistory table.
280011000	Cannot find StatusID for Quarantine in luUnitStatus table.
280012000	Cannot find StatusID for Processing in luUnitStatus table.
280012100	Invalid unit status in @xmlAllowedStatus.
280013000	Cannot find StatusID for Active in fsStatus table.
280013100	Invalid part status in @xmlAllowedStatus.
280014000	Cannot find StatusID for Active in luProductionOrderStatus table.
280015000	Cannot find StatusID for Released in luProductionOrderStatus table.
280015100	Invalid Production Order status in @xmlAllowedStatus.
280016000	Invalid unit.
280017000	Unit is not in an allowed status to run.
280017100	Unit is not in given status.
280018000	Unit does not have allowed part status.
280019000	The production order is not in active or released status.
280019100	The unit does have production order in allowed status.
280030010	Cannot pass in NULL as field name.
280030020	Cannot find column definition for field name.
280030050	Cannot find Unit in ffUnit table.
280030060	@ImpFieldName cannot be NULL.
280030062	More than one row is found in ffImportProduct for this unit.
280030066	No row is found in ffImportProduct for this unit.
280030067	Failed to read ffImportProduct table.
280030070	Data field is not defined as a column for ffUnitDetail.
280031010	Invalid UnitID being passed in.
280050010	Nonexistent UnitID passed into uspUNTGetSSN.
280050030	Nonexistent Serial Numnber Name passed into uspUNTGetSSN.
280050050	Unit does not exist in ffUnit table.
280050060	@SNName is not supplied.
280050070	@SNValue is not supplied.
280050080	Failed to insert luSerialNumber table.

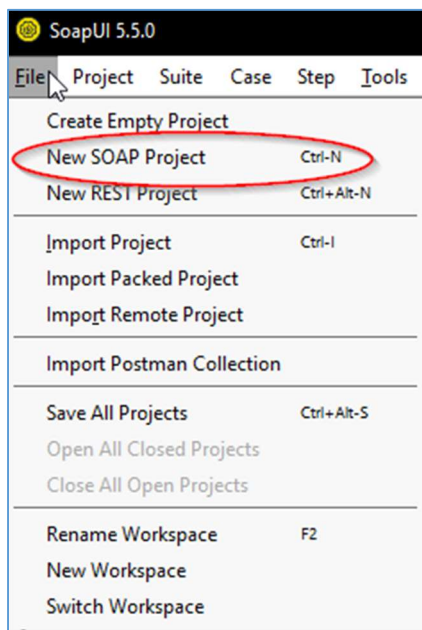
280050090	A serial number of given serial number type already exists for this unit.
280050100	Failed when update serial number data for unit.
280050110	Failed to insert ffSerialNumber table.
280060020	Cannot find unit in ffUnit.
280060030	Failed to generate serial number for ObtainValueBy=1.
280060040	Failed to call stored procedure for ObtainValueBy=5.
280060050	External stored procedure returned nonzero code.
280060060	ObtainValueBy falls out of range (1,2,3,4,5,6).
280060070	Failed to update ffUnitDetail table.
280060080	Cannot find ID in ffSNFormat table.
280060170	Failed to insert ffUnitDetail table [uspUNTGenField]
280060180	The newly generated field value is not the same as saved one in ffSerialNumber when verification is performed [uspUNTGenField]
280060190	The newly generated field value is not the same as saved one in ffUnitDetail when verification is performed [uspUNTGenField]
280070100	Failed to update ffUnit table.
280070101	Cannot find "Completed" status in luUnitStatus table.
280070105	Cannot find "Processing" status in luUnitStatus table.
280070200	Failed to insert ffHistory table.
280100130	SQL Server failed to update reserved_xx columns in ffUnitDetail table.
280100150	Some field name is not defined in fsFieldDefinition table.
280100190	Failed to insert into ffUnitDetail table.
280200050	When downgrade the partnumber for a unit, the target production order must be NULL.
280200100	Cannot find "Processing" in luUnitStatus table.
280200200	Cannot find production order StatusID for "released".
280200300	Cannot find production order StatusID for "active".
280200400	Invalid StationID.
280200500	Invalid ProductionOrderID.
280200600	Current production order must be in "active" or "released" status.
280200700	The new production order must define "IncomingPartNumber" in its detail table for part upgrade operation.
280200710	Cannot find PartID for IncomingPartNumber defined in ffProductionOrderDetail.
280200800	Unit does not exist in ffUnit table.
280200900	Unit cannot be assigned twice to the same production order.
280201000	Unit must have status 'Processing' or 'ReProcess' to be counted or moved to a production order.
280201100	Unit must have a part number.
280201200	Unit production order must be in "active" or "released" status when @DecrementOldPO is on.
280201300	The current production order containing unit must have an IncomingPartNumber.
280201310	Cannot find PartID for IncomingPartNumber defined in ffProductionOrderDetail.
280201400	The new production order must have an incoming part number when its PartID is not the same as that of the unit.
280201500	Unit's part number must match IncomingPartNumber of the new production order.

280201600	When downgrade unit's part number and assign it to a new PO, the new part number of the new PO must be consistent with that of current PO's incomingPartNumber.
280201800	The unit part number must be consistent with that of the new production order.
280202000	Failed to update ffUnit with new production order.
280202010	Failed to update ffUnitDetail with new production order.
280202100	Failed to update ffLineOrder table with new started quantity.
280202110	Failed to update ffLineOrder table with new started quantity.
280202200	The started quantity in ffLineOrder has reached limit, or the line order does not exist.
280202300	Failed to update ffLineOrder table with new started quantity.
280202310	Failed to update ffLineOrder table with new started quantity.
280202400	The started quantity of line order is zero, or the unit does not belong to a line order.
280221020	BTSAUTO must supply a serial number.
280221030	Multiple imatching PartID found for BTSAuto.
280221040	CANNOT find any matching PartID for BTSAuto.
600140100	Cannot find StationID for given station.
600140200	Cannot find @StationTypeID for given station.
600140300	Cannot find LineID for given station.
600140400	Cannot find default task group for nonexistent unit.
600140500	Cannot find task group "frmGetUnitInfo" based on existing unit info.
600140600	StartToPO is not defined for basic function frmGetUnitInfo.
600140700	UnitPattern is not defined for basic function frmGetUnitInfo.
600140800	ChangePartNumber is not defined for basic function frmGetUnitInfo.
600140850	SectionHeader is not defined for basic function frmGetUnitInfo.
600140870	Cannot find SectionHeader ID in ffSectionHeader table.
600140900	Unit serial number does not match UnitPattern.
600141000	Cannot find StationID for given station.
600141100	Cannot find @StationTypeID for given station.
600141200	Cannot find LineID for given station.
600141210	Invalid user ID.
600141220	The user is not allowed to enter this station.
600141300	Cannot find default task group for nonexistent unit.
600141400	Cannot find RouteID for default task group.
600141500	Cannot find task group "frmGetUnitInfo" based on existing unit info.
600141600	Cannot find next unit state ID for task group frmGetUnitInfo.
600141700	LineHasPO is not defined for basic function frmGetUnitInfo.
600141800	PNSource is not defined for basic function frmGetUnitInfo.
600141900	UnitPattern is not defined for basic function frmGetUnitInfo.
600142000	PNValidate is not defined for basic function frmGetUnitInfo.
600142100	SectionHeader is not defined for basic function frmGetUnitInfo.
600142200	ChangePartNumber is not defined for basic function frmGetUnitInfo.
600142250	AllowCreate is not defined for basic function frmGetUnitInfo.
600142300	Cannot find SectionHeader ID in ffSectionHeader table.
600142400	Unit serial number does not match UnitPattern.
600142500	Line must have a PO when part number is figured out from PO, or part number change is involved.
600142600	Cannot find PartID for case @PNSource=1.
600142700	Must indicate how to get part number when need create unit.

600142800	Part number of unit being created does not equal to that of the line PO's.
600142900	Unit part number is not the same as that the line PO's.
600143000	Part number of unit being created does not equal to IncomingPartNumber of the line PO.
600143100	Unit part number is not the same as the IncomingPartNumber of the line PO.
600143200	Part number of unit being created does not fall into PartID or IncomingPartNumber of the line PO.
600143300	Unit part number does not fall into PartID or IncomingPartNumber of the line PO.
600143350	Cannot create unit because flag AllowCreate is off.
600143400	Failed to find reverse part number when downgrade part number.
600143500	Failed to update ffUnit when downgrade part number.

8. SOAP UI Configuration

Soap UI is a tool that will help us to send and process our tests and visualize the response.



We will create a new SOAP project. In the Initial WSDL textbox we need set the *WSDL* end point of running service.

FFTesterService Service

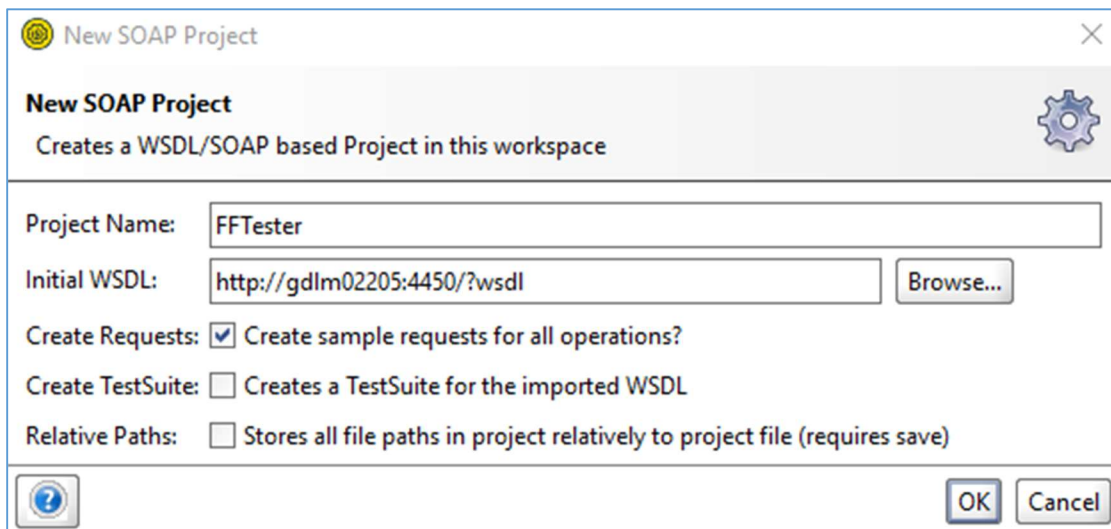
You have created a service.

To test this service, you will need to create a client and use it to call the service.

svcutil.exe <http://gdlm02205:4450/?wsdl>

You can also access the service description as a single file:

<http://gdlm02205:4450/?singleWsd1>

A dialog box titled "New SOAP Project" with a close button (X) in the top right corner. Below the title bar, there is a subtitle "Creates a WSDL/SOAP based Project in this workspace" and a gear icon. The main area contains several input fields and checkboxes: "Project Name:" with the text "FFTester" entered; "Initial WSDL:" with the text "http://gdlm02205:4450/?wsdl" entered and a "Browse..." button to its right; "Create Requests:" with a checked checkbox and the text "Create sample requests for all operations?"; "Create TestSuite:" with an unchecked checkbox and the text "Creates a TestSuite for the imported WSDL"; and "Relative Paths:" with an unchecked checkbox and the text "Stores all file paths in project relatively to project file (requires save)". At the bottom left is a help icon (question mark in a circle), and at the bottom right are "OK" and "Cancel" buttons.

New SOAP Project

Creates a WSDL/SOAP based Project in this workspace

Project Name:

Initial WSDL:

Create Requests: ☒ Create sample requests for all operations?

Create TestSuite: ☐ Creates a TestSuite for the imported WSDL

Relative Paths: ☐ Stores all file paths in project relatively to project file (requires save)

When the project has been created, we could observe our available functions .

